

1 Transport Property

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1.1 Boltzmann Transport Equation

Imagine there is a phase space, which is used to characterize the dynamics of one coherent state. Assuming the system is composed by a set of coherent state, with each approximately treated as nearly independent to each other. Then we have a distribution function, $g(\boldsymbol{\xi}, t)$, denotes the density of states at certain point. For Boson, the distribution obeys $g_0(\boldsymbol{\xi}) = \frac{1}{e^{\beta[H(\boldsymbol{\xi}) - \mu]} - 1}$. For Fermion, the distribution obeys $g_0(\boldsymbol{\xi}) = \frac{1}{e^{\beta[H(\boldsymbol{\xi}) - \mu]} + 1}$. Here μ is the chemical potential. For equilibrium, we have condition:

$$\frac{dg}{dt} = 0 \quad (1.1)$$

which gives:

$$\frac{\partial g}{\partial t} + \xi^i \frac{\partial g}{\partial \xi^i} = 0 \quad (1.2)$$

Replace the $\boldsymbol{\xi}$ with $\boldsymbol{x}$ and $\boldsymbol{k}$, we have:

$$\frac{\partial g}{\partial t} + x^i \frac{\partial g}{\partial x^i} + k^i \frac{\partial g}{\partial k^i} = 0 \quad (1.3)$$

For non-equilibrium state, we have equation $\frac{dg}{dt} = C[g]$, where $C[g]$ is the functional of g , stands for collision term. Under the relaxation time approximation (RTA), we have the equation:

$$C[g] \approx -\frac{g - g_0}{\tau} \quad (1.4)$$

So the Boltzmann transport equation now is:

$$\frac{\partial g}{\partial t} + x^i \frac{\partial g}{\partial x^i} + k^i \frac{\partial g}{\partial k^i} = -\frac{g - g_0}{\tau} \quad (1.5)$$

Under the linear response approximation, we can assume that, the density of states can be written as $g = g_0 + \delta g$, where δg is the small deviation from equilibrium.

The general Lagrange for coherent states is:

$$L = k_i \dot{x}^i - x_i \dot{k}^i - a_i \dot{x}^i - H \quad (1.6)$$